

Memo 07: Minimum-Effort Sediment Floc Simulation Using the Existing FVCOM Sediment Capability

FVCOM-MP WaterPACT Investigation

May 2026

Purpose

This memo records a code-level investigation of how to simulate sediment floc behavior by itself in the WaterPACT estuary setup, with minimum code and input disruption. The user request referred to `sed_mod.F`; in this repository there is no file with that exact name. The relevant sediment source files are:

- `mod_sed.F`: the original FVCOM sediment module, enabled by `-DSEDIMENT -DORIG_SED`.
- `mod_sed_cstms.F` and `mod_flocmod.F`: the CSTMS sediment and floc population-balance modules, enabled by `-DSEDIMENT -DCSTMS_SED`.

The practical question is whether we can run only sediment flocculation in the estuary without activating the microplastic floc-interaction machinery and without moving to a new sediment framework.

Main Finding

The minimum-effort path is to keep the original FVCOM sediment module (`ORIG_SED`) and change the sediment class from `non-cohesive` to `cohesive`. In `mod_sed.F`, the cohesive branch of `Calc_Deposition` automatically calls `Calc_Wset_Floc`; the non-cohesive branch uses the constant `SED_WSET`. Therefore, the current WaterPACT sediment files do not exercise the sediment floc settling logic because they set:

```
SED_TYPE = non-cohesive
```

Changing that one semantic switch, with a physically smaller `SED_SD50` and `SED_WSET`, gives a sediment-only floc experiment using existing code.

What the Existing ORIG Sediment Floc Path Does

For each cohesive sediment class, `mod_sed.F` computes a local settling velocity multiplier from suspended sediment concentration, salinity, turbulence intensity, and primary grain size:

$$W_s = W_{s0} \frac{a C^b S^d G^e}{(D_{50,\text{mm}})^f},$$

with default constants in the source:

$$a = 0.274, \quad b = 0.48, \quad d = 0.03, \quad e = 0.22, \quad f = 0.58.$$

The code enforces a minimum multiplier of 1.0 and only allows enhancement in brackish water:

$$5 < S < 15.$$

It also caps the settling velocity at $0.5\text{e-}3$ m/s, or 0.5 mm/s. This matters because the current WaterPACT sediment files use `SED_WSET = 3.0` mm/s for sand-like particles; if that value is reused for cohesive sediment, the result will immediately hit the cap. A flocc-only cohesive test should use a smaller base settling speed, for example `SED_WSET = 0.05--0.2` mm/s.

This path is lightweight but limited: it represents flocculation as a diagnostic settling-velocity adjustment, not as an explicit flocc size distribution.

Meaning of the Formula Variables

In the source routine `Calc_Wset_Floc(n,c,s,g1,ds50,w0,wset)`, the variables in the empirical multiplier are:

- **c**: suspended sediment concentration in the water column. In the ORIG sediment module this is stored internally as kg m^{-3} , which is numerically equal to gL^{-1} .
- **s**: salinity, from `s1(i,k)`.
- **g1**: a turbulence-intensity/shear proxy computed immediately before the call from local velocity, water depth, the Manning-like coefficient `mf=0.012`, gravity, and `Vcons=1.e-6`. It is not a direct GOTM/GLS dissipation rate; it is the ORIG module’s empirical turbulence-intensity argument for this formula.
- **ds50**: primary sediment diameter after conversion to meters. In the formula it is multiplied by 1000, so the denominator uses `D50` in mm.
- **w0**: base settling velocity without the flocc multiplier.
- **wset**: final settling velocity passed into the vertical settling calculation.

So, yes: in the shorthand interpretation of the formula, C is concentration, S is salinity, and G is the local turbulence/shear-intensity argument used by this empirical ORIG sediment closure.

Can This Be Used To Back Out an Equivalent Flocc Size?

Only with additional assumptions. The ORIG settling-velocity formulation does not prognose flocc diameter, flocc density, porosity, or fractal dimension. It returns an effective settling velocity. A unique flocc size cannot be recovered from settling velocity alone because flocc settling depends on both diameter and effective density, and flocc effective density itself usually varies with flocc size and composition.

For post-processing, one could compute a diagnostic “Stokes-equivalent” diameter by assuming a spherical flocc with a specified effective density:

$$d_{\text{eq}} = \left(\frac{18\mu W_s}{g(\rho_f - \rho_w)} \right)^{1/2}.$$

This is useful only as an interpretation aid. It is sensitive to the assumed $\rho_f - \rho_w$, and it becomes especially ambiguous near the hard cap $W_s \leq 0.5 \text{ mm s}^{-1}$.

A more floc-aware diagnostic would assume a primary particle diameter d_p , sediment density ρ_s , water density ρ_w , and fractal dimension n_f , for example:

$$\rho_f - \rho_w = (\rho_s - \rho_w) \left(\frac{d_p}{d_f} \right)^{3-n_f}.$$

Under Stokes-flow assumptions this gives:

$$d_f = \left[\frac{18\mu W_s}{g(\rho_s - \rho_w)d_p^{3-n_f}} \right]^{1/(n_f-1)}.$$

That calculation would still be diagnostic, not part of the current model state. If explicit floc size is a primary output requirement, the CSTMS `mod_flocmod.F` route is the more appropriate model structure.

Planned MP–Floc Coupling Extension

The preferred future coupling plan is to keep the simplified ORIG sediment floc module internally unchanged, while using a diagnostic/effective floc size only inside the plastic–floc interaction pathway. In other words:

- `SED_SD50` remains the constant primary cohesive sediment size used by `mod_sed.F`.
- The sediment floc formula continues to compute the local floc-affected settling velocity and stores it internally as `sed(ised)%wset0(i,k)`.
- `plast(iplast)%dequ` remains the true MP equivalent particle diameter. It should not be overwritten by a floc diameter.
- A separate diagnostic variable, conceptually `d_floc_eff(i,k,ised)`, would represent the effective floc size used only for MP–floc aggregation, size-exclusion, and related plastic interaction terms.

This is a one-way coupling approximation: sediment concentration, salinity, turbulence, and primary sediment size determine the floc environment; MP then responds to that environment. Because MP concentration is much smaller than suspended sediment concentration in the intended application, MP does not need to feed back on the sediment floc state in this minimum implementation.

Current Behavior to Preserve

In the present `mod_plast.F` floc-interaction pathway, the switch `PLAST_FLOC` activates the two-tracer aggregated/disaggregated MP logic. The current implementation uses sediment D50-based proxies:

- the MP–floc aggregation kernel uses `sed(ised)%Sd50` as the floc-size proxy;
- the differential-settling term uses the class settling velocity `sed(ised)%wset`;
- aggregated MP settling currently uses the sediment settling velocity proxy already in the code.

This current behavior should remain exactly available as the baseline option.

Proposed New Switch

Add one additional plastic-module runtime switch after the literature parameterization is selected, for example:

```
PLAST_FLOC_EFFECTIVE = F
```

The exact name can be chosen during implementation. The required behavior is:

- **When false:** preserve the current `mod_plast.F` behavior exactly. MP–floc interaction uses sediment `Sd50/Wset` proxies, with no diagnostic effective floc size.
- **When true:** use the modified floc quantities only inside the plastic–floc interaction pathway. The aggregation and size-exclusion calculations should use `d_floc_eff(i,k,ised)`, and the aggregated MP settling calculation should use the floc-affected sediment settling velocity, preferably `sed(ised)%wset0(i,k)` or a sediment-concentration-weighted equivalent if multiple cohesive classes are active.

The open science item is the parameterization for `d_floc_eff`. The simple Stokes or fractal inversion written above is not yet satisfactory as a production closure. Before implementing this switch, we should choose a defensible literature-based relation for effective floc diameter and, if needed, floc effective density/porosity. That reference choice should also define how sediment mass concentration is converted to floc number density in the MP aggregation kernel.

Reference Trail for the ORIG Formula

The immediate source-code reference is the comment block above `Calc_Wset_Floc`. It attributes the cohesive floc settling method to Qingyu Sha and lists:

- Ding P.X., Hu K.L., Kong Y.Z., Zhu S.X. et al. (2003). “Numerical simulation of total sediment under waves and currents in the Changjiang Estuary.” *Acta Oceanologica Sinica*, 25(5):113–124. The same citation is visible in online bibliographic lists and environmental reports as a Chinese article with an English abstract.
- Hu K.L. (2003). “2-D Suspended Sediment Numerical Simulation under Waves and Currents in Yangtze Estuary.” Ph.D. dissertation, East China Normal University, as cited by the source comment.

I could verify the bibliographic identity of the Ding et al. paper online, but I did not find an openly accessible copy containing the exact empirical coefficient equation. The strongest citation for the exact coefficient set is therefore the FVCOM source comment itself, which appears to have embedded the Changjiang/Yangtze-estuary empirical relation directly from the cited Chinese paper/dissertation trail.

Online sources checked include a later Yangtze Estuary process-modeling paper whose reference list includes Ding et al. (2003), and a Changjiang Estuary flocculation review/survey PDF. Those sources support the same physical dependencies used by the code: concentration, salinity, particle size, and hydrodynamic/turbulence conditions control floc formation and settling. Recent summaries also place an important Changjiang flocculation salinity range around 5–10, which is consistent with the source code’s brackish activation gate of 5–15.

- Process-modeling reference list checked: https://www.researchgate.net/publication/292835947_A_process-based_approach_to_sediment_transport_in_the_Yangtze_Estuary
- Changjiang Estuary flocculation survey PDF checked: https://zjjcmspublic.oss-cn-hangzhou-zwynet-d01-a.internet.cloud.zj.gov.cn/jcms_files/jcms1/web2548/site/attach/0/15fb3123b5794fea9144293da397749a.pdf

Current WaterPACT Configuration Relevant to Floc-Only Runs

Table 1: Current input features that matter for a sediment-only floc test.

Item	Current setting	Implication
Run namelist	<code>SEDIMENT_MODEL = T</code>	Sediment runtime switch is already used.
Run namelist	<code>PLASTIC_MODEL = T</code>	Turn this off for sediment-only floc.
Sediment file	<code>SED_TYPE = non-cohesive</code>	Floc settling path is inactive.
Sediment file	<code>SED_WSET = 3.0 mm/s</code>	Too large for cohesive floc branch.
Sediment file	<code>MORPHO_FACTOR = 10.0</code>	Accelerates bed exchange; use 1.0 for cleaner physics.
Sediment file	<code>INIT_BED_FRACTION</code> has six entries	For <code>NSED=1</code> , set it to 1.0.
River file	variable <code>coarse_sand_1</code> exists	Keep this sediment name to avoid NetCDF edits.

The existing river NetCDF file `waterPACT_riv_floc_MP.nc` already contains a sediment concentration variable named `coarse_sand_1`, with values about $4.5\text{e-}4$ to $1.6\text{e-}2$ g/L. FVCOM looks up river sediment variables by the sediment class names passed from `SETUP_SED`; therefore, the lowest-effort route is to keep `SED_NAME = coarse_sand_1` even if the class is reinterpreted as cohesive mud/floc material. If the name is changed to `mud_floc` or similar, the river NetCDF file must also contain a matching variable.

Recommended Minimum-Effort Setup

1. Use the ORIG sediment build

For a fresh compile, make sure the executable is built with the original sediment module:

```
FLAG_21  = -DSEDIMENT
FLAG_211 = -DORIG_SED
```

The checked `make.inc` currently has these sediment flags commented, while `FLAG_49 = -DPLASTIC` is enabled. Existing executables may have been compiled from a different flag state, so this should be confirmed before rebuilding or submitting a new production run.

2. Turn off plastic in the run namelist

Clone one of the current WaterPACT run directories and change only the model switches needed to remove microplastic:

```

SEDIMENT_MODEL          = T,
SEDIMENT_MODEL_FILE      = 'generic_sediment_floc_only.inp',
SEDIMENT_PARAMETER_TYPE  = 'uniform',
SEDIMENT_PARAMETER_FILE  = 'none',
PLASTIC_MODEL            = F,
PLASTIC_MODEL_FILE       = 'none'

```

Keep the existing river forcing:

```

RIVER_NUMBER            = 9,
RIVER_INFO_FILE         = 'waterPACT_riv_floc_MP.nml',
RIVER_KIND              = 'variable',
RIVER_TS_SETTING        = 'calculated',
RIVER_INFLOW_LOCATION   = 'node'

```

The river file contains both sediment and plastic variables. With `PLASTIC_MODEL = F`, the plastic variables are simply irrelevant.

3. Create a cohesive sediment input file

Copy `generic_sediment_c.inp` or `mp_sediment_a_no_adv.inp` to `generic_sediment_floc_only.inp`. Then make these key edits:

```

BEDLOAD = F
SUSLOAD = T
SED_PTSOURCE = T
MORPHO_MODEL = F
MORPHO_FACTOR = 1.0

NSED = 1
SED_NAME = coarse_sand_1
SED_TYPE = cohesive
SED_SD50 = 0.004
SED_WSET = 0.1
SED_SRHO = 2650.
SED_ERAT = 0.58E-3
SED_TAUE = 0.1
SED_TAUD = 0.1
SED_PORS = 0.5

NBED = 1
INF_BED = T
INIT_BED_POROSITY = 0.50
INIT_BED_THICKNESS = 10.0
INIT_BED_FRACTION = 1.0

```

`SED_SD50 = 0.004 mm` is a placeholder primary-particle scale consistent with the floc literature values used in `mod.flocmod.F`. For Delaware/Schuylkill suspended sediment, this should be tuned to the available grain-size data. The important code-facing point is that it should be much smaller than the current sand values 0.1--1.0 mm.

What Not To Expect From This Minimum Path

- It will not output an explicit floc diameter or full floc size distribution. The source only stores the floc-affected settling velocity internally in `sed(i)%wset0`; that field is not registered as a standard sediment NetCDF output in the current `mod_ncdio.F`.
- It will not use `SED_CSED` for initial water-column sediment. The `ORIG` initialization sets suspended sediment concentration to zero; the practical sediment source is the river forcing unless `init_sed.F` is edited.
- In the cohesive settling flux routine, the critical deposition stress gate is commented out, so `SED_TAUD` should not be relied on to suppress deposition under high shear in this path.
- The floc enhancement activates only where the simulated salinity is between 5 and 15 PSU. River-source salinity in the forcing file is zero, so floc enhancement is expected mainly in the brackish mixing region, not at the fresh river nodes.

Why Not Use `mod_flocmod.F` First?

`mod_flocmod.F` is the more complete sediment floc model. It computes transfers among cohesive sediment classes through shear aggregation, differential-settling aggregation, shear fragmentation, optional collision-induced fragmentation, and optional bed deflocculation. However, it is only active in the CSTMS build:

```
#if defined (SEDIMENT) && (CSTMS_SED)
```

and is called from `mod_sed_cstms.F` only when `SED_FLOCS = T`. Using it would require:

- rebuilding with `-DCSTMS_SED` instead of `-DORIG_SED`;
- writing a CSTMS-format sediment input with `NCS`, `NNS`, `MUD_*`, `SED_FLOCS`, and `f_*` parameters;
- creating river NetCDF variables for every cohesive class name;
- rechecking NetCDF output and restart behavior for the CSTMS variables.

That is a valid future path if the research question requires an evolving floc size distribution. It is not the minimum-effort path for a first estuary-scale sediment-floc-only experiment.

Recommended First Experiment

Run a short pair of tests using the same hydrodynamics and river sediment forcing:

- (1) **No-floc control:** one sediment class with the same `SED_SD50` and `SED_WSET`, but `SED_TYPE = non-cohesive`.
- (2) **Floc case:** identical file except `SED_TYPE = cohesive`.

Compare the output variable `coarse_sand_1` in surface, bottom, and depth-averaged views, plus `coarse_sand_1.bedfrac`. This isolates the effect of the `ORIG` floc-settling formula without invoking plastic aggregation or the larger CSTMS framework.

Bottom Line

For minimum effort, do not modify source code and do not start with `mod_flocmod.F`. Use the existing `ORIG_SED` module, turn off plastic, keep the existing river sediment variable name, and run one cohesive sediment class. This gives a sediment-only flocc sensitivity experiment immediately. Use CSTMS later only if the goal changes from “floc-affected settling” to an explicit evolving flocc population.